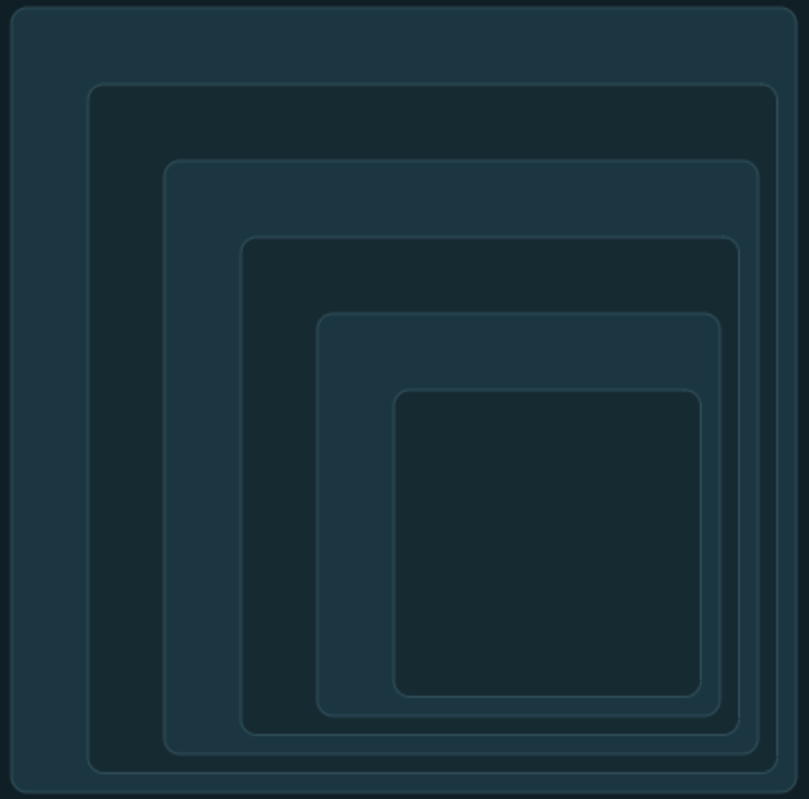
DIGITAL MINE ON OPENGIZ

Ademix

3D open-pit operations twin

A connected haulage and crushing simulation built for decisions, not decoration.



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One disruption changes the whole production chain

Dispatchers need to see consequences across equipment, roads, stockpiles and the crusher in one place.

01 **Observe** — Live state for trucks, excavator, road, ROM pad and crusher

02 **Intervene** — Trigger breakdown, road closure or crusher slowdown

03 **Compare** — Read projected throughput against the 900 t/h baseline

DECISION LOOP

OpenEgiz is the source of truth

The browser does not invent mine state. Telemetry flows through MQTT into Eclipse Ditto twins, then the interface reads it through the API.



Eleven twins describe one connected mine

Relationships make the model navigable. Features carry operational state, not only geometry.

Mine system

- **North Pit**
- **Haul road**
- ROM stockpile

Mobile fleet

- **3 haul trucks**
- **1 excavator**
- Position • speed • payload • status

Processing + control

- **Crusher**
- **Scenario controller**
- Inventory • processed tonnes • constraints

Twin relationships mine → assets → process state → scenario impact

The 3D scene is an operational interface



Conceptual interface composition

Interface layers

Spatial

Camera, benches, route, animated fleet

Operational

KPIs, asset selection, status and payload

Decision

Scenario controls and projected impact

Mass balance keeps the simulation credible

Truck cycles deliver ore to the ROM pad. The crusher consumes inventory continuously, so material is never deleted at dump time.

77 t
truck payload

250 t
initial ROM inventory

900 t/h
baseline crusher rate

15.5 km/h
closed-road speed

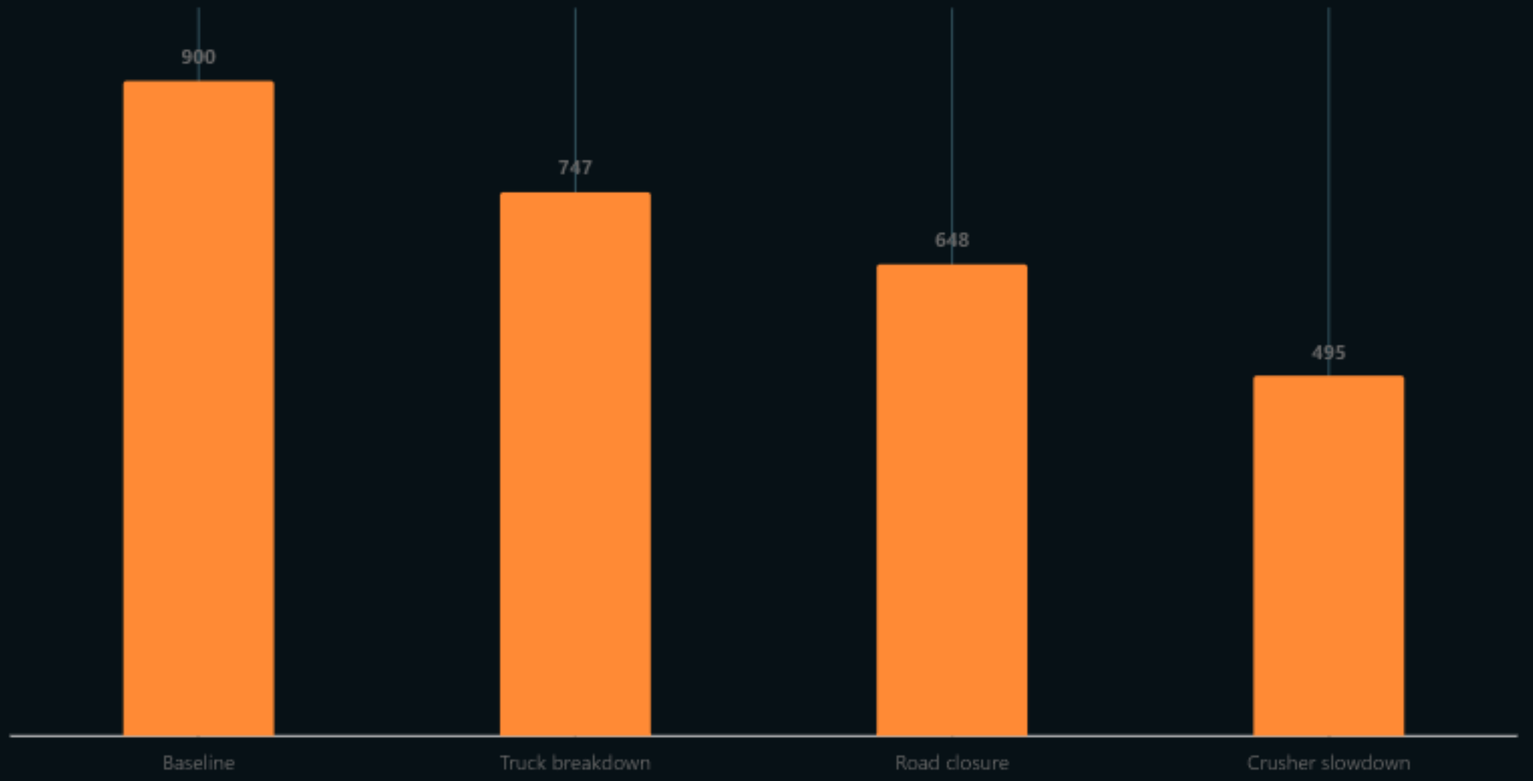
Material accounting

initial ROM + delivered ore = current ROM + processed ore

Scenario parameters are explicit, testable and visible to the jury.

What-if modes quantify operational impact

Projected throughput uses the same baseline across four deterministic modes.



Impact from baseline

Truck breakdown	−17%
Road closure	−28%
Crusher slowdown	−45%

Evidence is separated from release gates

What is already verified is stated plainly. Remaining environment checks stay visible until they pass.

Verified

- **8 simulation unit tests pass**
- Frontend production build passes
- Python compile and JSON validation pass
- Compose configuration parses

Before submission

- **Run clean end-to-end stack**
- Capture real 3D interface evidence
- Record demo and rehearse jury flow
- Confirm final commit on official main

NO FABRICATED TELEMETRY • NO HIDDEN CLAIMS

Ademix turns mine events into decisions

Connected twins. Visible material flow. Measurable scenarios.

1 Select an asset

2 Trigger a scenario

3 Compare throughput